Continuum limit of the perturbative Brazovskii correlators – program and open lemma

TECT verification-first repository · claim A3-PERTURBATIVE-CONTINUUM-CORRELATORS

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1. Purpose and scope

This note sets up the continuum limit of the *perturbative correlation functions* of the scalar Brazovskii theory and isolates the single open lemma that separates it from a theorem. It is a PROOF SKETCH (T3): the strategy is written, two gaps are marked OPEN. Scope: scalar, $d = 3$, $\mu^2 > 0$, perturbative; NOT constructive, NOT full Class-II, NOT $\mu^2 < 0$. The UV-finiteness input (the uniform q^{-4} bound and absolute convergence) is the proved sibling `A3-UV-SUPERRENORMALISABILITY`.

2. The perturbative measure

UV-finiteness (A3-UV) concerns Feynman amplitudes; it does not by itself define correlators. Fix a regularised theory by a lattice spacing a and a momentum cutoff Λ, with regulated kernel K_a and action $F_{\Lambda,a}$. Define

$$d\nu_{\Lambda,a}(\phi) = Z_{\Lambda,a}^{-1}\, e^{-F_{\Lambda,a}[\phi]}\, D\phi, \qquad \langle \mathcal{O} \rangle_{\Lambda,a} = \int \mathcal{O}(\phi)\, d\nu_{\Lambda,a}(\phi), \tag{1}$$

with $Z_{\Lambda,a}$ the partition function. The perturbative expansion of $\langle \mathcal{O} \rangle_{\Lambda,a}$ is a sum over connected graphs $\mathcal{G}$ with amplitudes $\mathcal{A}_{\mathcal{G},a}(p_1, \ldots, p_n)$.

3. The graphwise convergence lemma (target) and strategy

Lemma (A3-GRAPHWISE-CONVERGENCE, OPEN). *For a suitable regulator family $K_a \to K$ and every connected graph $\mathcal{G}$,*

$$\lim_{a\to 0} \mathcal{A}_{\mathcal{G},a}(p_1, \ldots, p_n) = \mathcal{A}_{\mathcal{G}}(p_1, \ldots, p_n), \tag{2}$$

uniformly on compact external momenta, with the $a \to 0$ and $\Lambda \to \infty$ limits agreeing.

Strategy (dominated convergence). (i) the lattice propagator converges pointwise; (ii) it obeys the uniform q^{-4} UV bound of A3-UV; (iii) by the Weinberg condition every subintegral is absolutely convergent, giving an integrable dominating function; (iv) dominated convergence then exchanges the limit and the loop integrations for each graph. UV-finiteness (A3-UV) supplies exactly the uniform bound and absolute convergence that the dominating function requires; the single-graph instance (the tadpole, with $\mathcal{A}_{\mathcal{G},a} \to \mathcal{A}_{\mathcal{G}}$ to 10^{-12}) is verified in `codes/foundations/a3_renormalisation_checks.py`.

4. Open gaps

OPEN GAP 1 (the lemma). The dominated-convergence argument above is written for a generic graph but not executed: the uniform integrable dominating function must be constructed for arbitrary nested loop structures, and the pointwise convergence $G_a \to G$ proved for the chosen K_a.

OPEN GAP 2 (regulator matching). The regulator family K_a and the matching of the Brillouin-zone cutoff to the continuum kernel must be fixed so that $a \to 0$ and $\Lambda \to \infty$ are the *same* limit; this is a definition that the present note states but does not pin down.

Closing both gaps promotes this claim to T6 jointly with A3-UV (scalar/d=3/$\mu^2 > 0$ perturbative scope).

5. Devil's-advocate (three objections)

(α) *"A3-UV already gives this."* DISMISSED: A3-UV gives per-graph UV-finiteness at fixed regularisation; it does not give the $a \to 0$ *limit* of the correlators, which needs the measure and the graphwise lemma.

(β) *"The tadpole instance suffices."* DISMISSED: one convergent graph does not prove convergence of all graphs; the lemma is quantified over every connected $\mathcal{G}$.

(γ) *"$a \to 0$ and $\Lambda \to \infty$ are obviously the same."* UPHELD as a gap: they coincide only for a specified regulator/cutoff matching (Gap 2); asserting it without that definition is unjustified.

Quantitative sanity check (mandatory): the uniform UV bound exponent is the q^{-4} of A3-UV (verified $K(q)/(Yq^4) \to 1$); the one-graph dominated-convergence instance (tadpole) converges to 10^{-12} on refinement; the general lemma's status is OPEN (no numerical claim is made for arbitrary graphs).

6. Result footer

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Result ID:                A3-PERTURBATIVE-CONTINUUM-CORRELATORS
Precise statement:        PROOF SKETCH: with dnu=Z^-1 e^{-F} Dphi,
                          all connected perturbative correlators
                          -> a cutoff-independent continuum limit
                          as a->0, Lambda->inf, CONDITIONAL on the
                          open graphwise lemma lim_a A_{G,a}=A_G
                          (A3-GRAPHWISE-CONVERGENCE).
Scope:                    scalar, d=3, mu^2>0, perturbative;
                          PROOF SKETCH.
Dependencies:             A1-KERNEL-CONV;
                          A3-UV-SUPERRENORMALISABILITY.
Open gaps:                (1) graphwise dominated-convergence
                          lemma; (2) K_a regulator / BZ matching.
Evidence grade:           ANALYTIC (strategy) + ESTIMATOR
                          (tadpole only).
Reproduction command:     n/a (open); tadpole instance in
                          a3_renormalisation_checks.py.
Falsification gate:       a connected graph G with A_{G,a} not->
                          A_G; or non-uniqueness of a->0 vs
                          Lambda->inf.
Tier before / after:      new / T3 PROOF SKETCH.
No-overclaim statement:   Not a theorem; one tadpole instance
                          does not prove the all-graph lemma; not
                          constructive; not Class-II; not mu^2<0.
Next required action:     Prove A3-GRAPHWISE-CONVERGENCE + fix the
                          regulator matching; then T6 with A3-UV.
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